

RECOVERY STATUS OF SCLERACTINIAN CORALS AND ASSOCIATED FAUNA IN THE ANDAMAN AND NICOBAR ISLANDS

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ABSTRACT: Percentage coral cover, number of species and colonies of hermatypic corals and associated fauna in each of the 20 sites were recorded throughout the year from July 2009 to March 2011. Percentage coral cover was evaluated from LIT (Line Intercept Transect) data as well as underwater photography at each of the study sites. Evaluation of the collected data showed that the maximum live coral cover was at North Andaman (17.5 %) followed by the Nicobar Islands (15.4 %), Middle Andamans (14.8 %) and South Andaman (14 %). Compared to the 2010 data, maximum recovery of live corals was reported at North Andaman (7 %) and minimum in Nicobar regions (0.6 %). Maximum bleaching was reported in North, South and Middle Andaman Islands whereas least bleaching was noted in the Nicobar Islands.

INTRODUCTION

The Andaman and Nicobar Islands, located in the Bay of Bengal between latitudes 6° 45' N to 13° 41' N and longitudes 92° 12' E to 93° 57' E, are rich in biodiversity. They are one of the few key biodiversity regions of the world surrounded by fringing coral reefs characteristic of the Southeast Asian region, and are the most diverse reef areas of the Indian subcontinent (Pillai, 1983). They have an area of about 12,000 sq. km. consisting of reef lagoons, coral banks, reef slopes and reef flats (Wilkinson, 2002) of which, an area of 733.12 sq. km. has been designated as 'protected area' under the Wild Life (Protection) Act of 1972 (Dorairaj and Soundararajan, 1987).

Coral reefs provide food, building material and shoreline protection for people living near them and also promote tourism. They typically grow in relatively clear waters. The health of coral reefs deteriorated recently throughout the world due to various natural and anthropogenic causes (Hughes, 1994; Wilkinson, 1998; 2000; 2002; 2004; Bellwood *et al.*, 2004). Land use practices such as overgrazing as well as industrial and coastal development increase the supply of sediment to near shore areas where most coral reefs develop, creating stressful conditions that may ultimately lead to the death of corals (Rogers, 1990; Buddemeier and Hopley, 1998; Fortes, 2000).

High concentrations of suspended sediment can reduce the coral growth rate (Dodge *et al.*, 1974; Dodge and Vaisnys, 1977; Hubbard and Scature, 1985; Edmunds and Spencer-Davies, 1989), decrease net productivity (Rogers, 1979; Edmunds and Spencer-Davies, 1989) and induce coral stress (Van Katwijk *et al.*, 1993). Suspended sediments can also be a vector for the introduction of nutrients, toxic substances and heavy metals into reefs that can cause stress or kill corals (Dickson *et al.*, 1987; Souter *et al.*, 1993; Bastidas *et al.*, 1999). Climatic changes lead to an increase in the sea surface temperature, a critical factor affecting the well-being of the symbiotic association between host animals like hard corals, giant clams, soft corals and sea anemones with algal symbionts (Buddemeier *et al.*, 2004). Although corals can often recover from bleaching, they will die if the stress is extreme or prolonged (Wilkinson, 1998).

In this study, the percentage of coral cover, number of species and colonies of hermatypic corals and associated fauna in each site were recorded throughout the year from July 2009 to March 2011. Percentage coral cover was evaluated from LIT (Line Intercept Transect) data as well as underwater photography at each of the study areas. During the survey, changes were observed in the hermatypic corals and associated fauna in each site. Based on the data obtained, the

bleaching and recovery status of corals at all the study areas are presented.

MATERIALS AND METHODS

Coral reef communities were monitored at four regions of the Andaman and Nicobar Archipelago *i.e.* North, South and Middle Andaman as well as the Nicobar Islands. Twenty permanent monitoring sites covering a total quadrat area of 500 m² were selected in each of the study islands under the National Coral Reef Research Institute (NCRI) project from July 2009 till date (Table.1). Quadrats having a minimum area of 1 m² were divided into a uniform grid of 100 segments as followed by Greig-Smith (1983). The survey was

carried out by SCUBA diving at depths ranging between 5 m to 30 m. Transects of 20 m length were laid along the reef area at permanent monitoring plots and depending on the size of the reefs, 8 to 12 transects were laid at each study site of each island. The percentage cover of each life form category was then calculated following the method of English *et al.* (1997). The difference between the data obtained in 2010 and 2009 gave the percentage of bleaching mortality and the difference between the data obtained in 2011 and 2009 gave the percentage of recovery at the permanent monitoring sites of each island.

$$\text{Percentage cover} = \frac{\text{Total length of category}}{\text{Length of transect}} \times 100$$

Table 1. Permanent monitoring sites in the Andaman and Nicobar Islands

S.No	Study sites	Geographical location
North Andaman		
1	Ross Island	Lat. 13°18.167' N; Long. 93°04.261' E
2	Smith Island	Lat. 13°18.406' N; Long. 93°04.207' E
3	Ariel Island	Lat. 13°16.093' N; Long. 93°02.433' E
4	Lamia Island	Lat. 13°12.879' N; Long. 93°05.516' E
5	North Reef Island	Lat. 12°56.084' N; Long. 92°57.345' E
South Andaman		
6	Rut land Island	Lat. 11°29.456' N; Long. 92°36.889' E
7	Tarmugli Island	Lat. 11°34.138' N; Long. 92°33.836' E
8	Bell Island	Lat. 11°34.095' N; Long. 92°33.841' E
9	Chester Island	Lat. 11°35.233' N; Long. 92°34.688' E
10	Twins Island	Lat. 11°23.773' N; Long. 92°33.097' E
Middle Andaman		
11	Quaiter Island	Lat. 12°20.323' N; Long. 92°54.529' E
12	Long Island	Lat. 12°21.749' N; Long. 92°55.410' E
13	North Passage	Lat. 12°18.121' N; Long. 92°55.718' E
14	Karmatang	Lat. 12°51.322' N; Long. 92°56.050' E
15	Interview Island	Lat. 12°59.125' N; Long. 92°42.981' E
Nicobar islands		
16	Champion Island	Lat. 08°01.718' N; Long. 93°32.728' E
17	Trinket Island	Lat. 08°03.352' N; Long. 93°34.102' E
18	Kamorta	Lat. 08°02.145' N; Long. 93°33.117' E
19	Katchal island	Lat. 08°00.002' N; Long. 93°24.362' E
20	Munak island	Lat. 07°59.806' N; Long. 93°29.852' E
21	Car Nicobar	Lat. 09°08.310' N; Long. 92°49.879' E

RESULTS

The results of the line intercept transect studies conducted from July 2009 to March 2011 for all the 20 permanent monitoring stations in the four regions (North, South and Middle Andaman, and Nicobar) were as follows:

As in the 2009 survey, North Andaman had the highest percentage of live coral, dead coral and dead coral with algae; Middle Andaman was rich in rubble and sponges; South Andaman was rich in other associated fauna and the Nicobar Islands had abundant dead coral (Table 2, Fig. 1).

During the period of coral bleaching (March–June 2010), a rapid survey was carried out. The corals of South Andaman were the most

bleached (both partially and completely); followed by that of North Andaman, Middle Andaman and the Nicobar Islands (Table 3, Fig. 2).

The 2011 (January–March) results showed that maximum live coral cover was at North Andaman followed by the Nicobar Islands, Middle Andaman and South Andaman. There appeared to be some recovery of live corals at North Andaman but very limited recovery in the Nicobar Islands (Table 4, Fig. 4).

DISCUSSION

There was serious coral bleaching in the Andaman and Nicobar Archipelago in 2010 and



Figure 1. Live coral cover at different study sites before bleaching (July 2009–Feb 2010)

A-LIT method coral reef survey and Photo Documentation

B-*Pocillopora woodjonesi* Vaughan, 1918

C-*Acropora divaricata* (Dana, 1846)

D- *Seriatopora hystrix* Dana, 1846

by 2011 most of the corals were dead and some of the dead corals were covered with algae (Fig. 3).

Earlier, severe coral bleaching (65% to 80%) in the Andaman and Nicobar Islands had taken place in 1998 during the months of May and June due to the rise in seawater surface temperature (Pet-Soede *et al.*, 2000). After that in 2002 and 2005, 20 to 40% of corals in these islands were bleached as indicated by the reports of Soundararajan (2002) and Damroy *et al.* (2006). The highest live cover at this time was reported in the Nicobars (73.5%) followed by Outram Island in South Andaman (67.3%) and North reef in North Andaman (72.7%) (Rajasuriya *et al.*, 2002). Subsequently the bleaching status of corals in 2010 was reported for the South Andaman region (Krishnan *et al.*, 2011)

Compared to previous reports, the bleaching event of 2010 was the most severe ever

reported from the Andaman and Nicobar Islands as shown by our LIT data (Figs. 1-4). Presently corals show the highest maximum live coral cover at North Andaman (17.5%) followed by the Nicobar Islands (15.4 %), Middle Andamans (14.8%) and South Andaman (14.3 %). In 2011 maximum recovery of live corals was reported at North Andaman (7%) and least recovery in the Nicobar Islands (0.6%). Maximum bleaching was reported at North, South and Middle Andaman while fewer corals bleached in the Nicobars. Coral mortality may be the result of the stresses of high temperatures, and the combined stresses of high irradiance, salinity changes, turbid water and exposure during extreme low tides. The levels of temperature stress experienced by corals described in this study exceed the values known to cause major coral mortality (Coles and Jokiel, 1978; Brown, 1997; Hoegh-Guldberg, 1999; Strong *et al.*, 2000; IPCC, 2001; WWF, 2004).

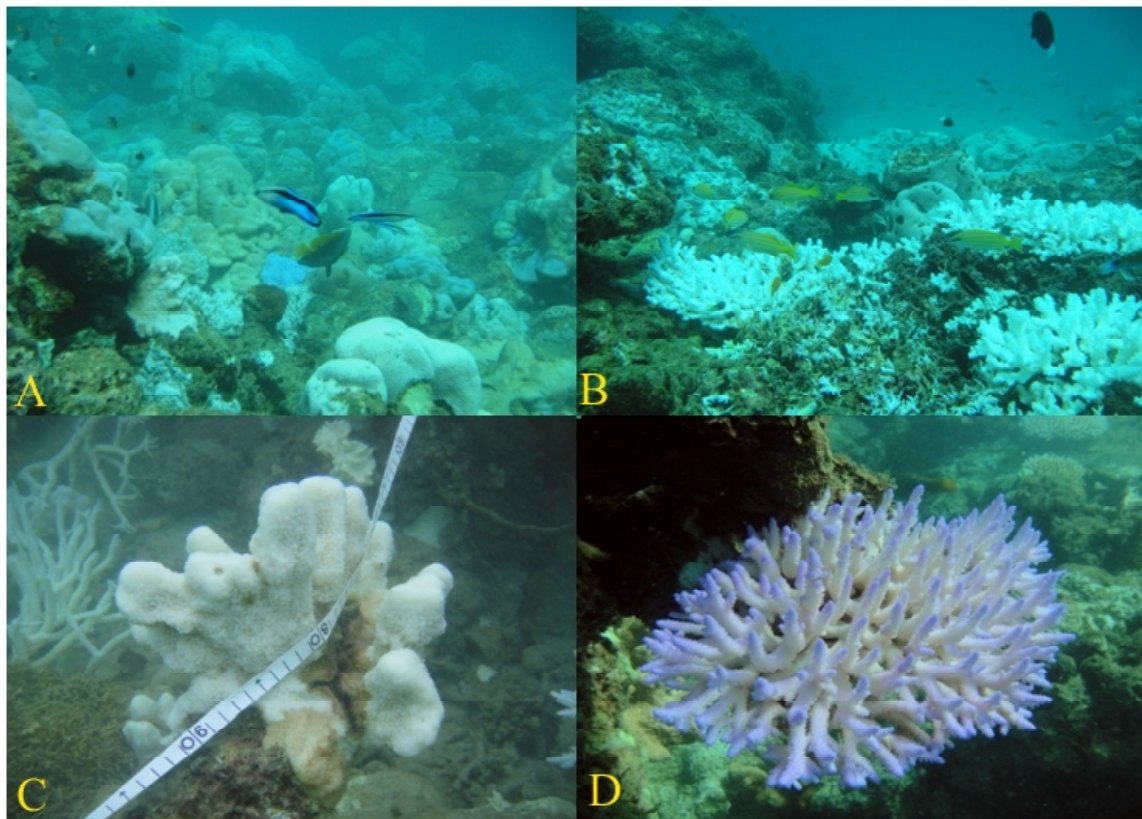


Figure 2. Coral cover at different study sites during bleaching (March–June 2010)

A-Bleached massive coral reef beds

B-Bleached branching coral reef beds

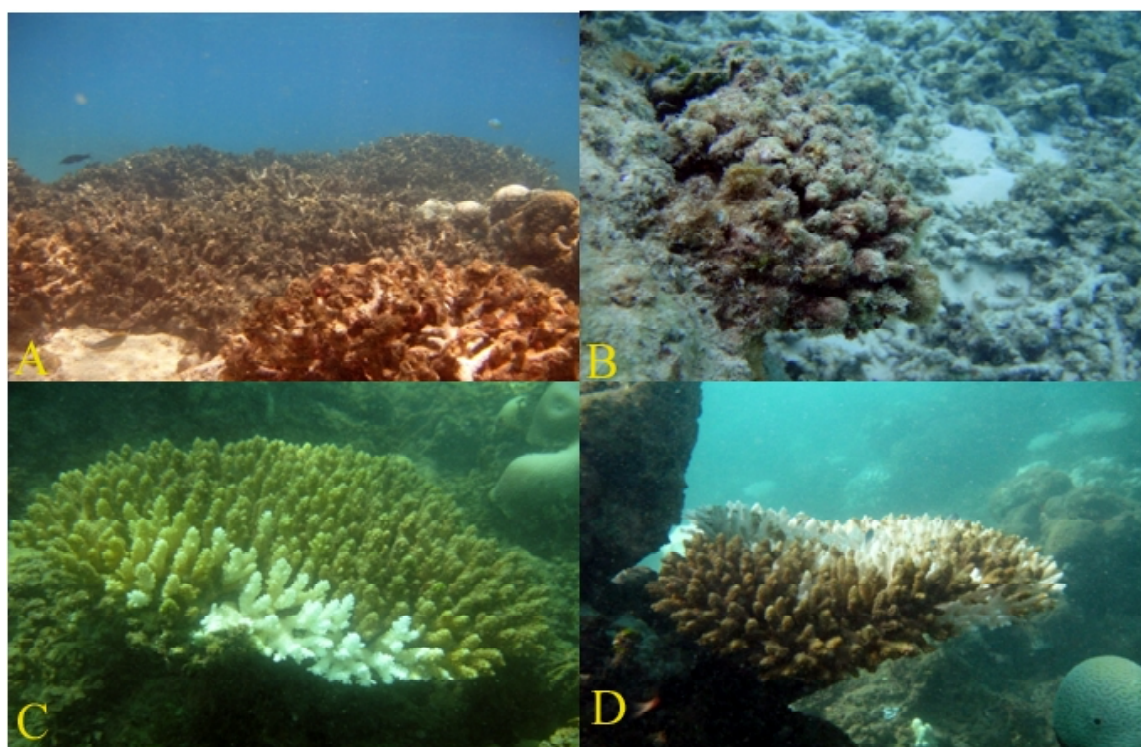
C- *Acropora palifera* (Lamarck, 1816)

D- *Acropora valenciennesi* (Milne Edwards and Haime, 1860)

Table 2. Percentage of live corals, rubble and associated fauna in 2009–2011 at different survey sites in the Andaman and Nicobar Islands.

% of Cover	North Andaman			South Andaman			Middle Andaman			Nicobar		
	2009	2010	2011	2009	2010	2011	2009	2010	2011	2009	2010	2011
LC	48.3	10.5	17.5	42.6	11.5	14.8	39.8	11.6	14.3	29.8	14.8	15.4
PB	0	17.2	9.4	0	17.3	12.3	0	13.3	13.1	0	12.2	11.4
CB	0	11.6	10.3	0	13.1	10.9	0	9.1	11.1	0	16	10
SC	7.9	7.8	5.1	7	7.2	6.2	6.6	7.9	6.5	4.3	5.9	6.1
DC	5.2	14.9	20.3	13	14	17.9	14.6	14.6	16.6	21.7	17.2	17.7
DCA	4.6	4.6	9.9	3.3	4.2	8.5	3.1	5.4	6.7	1.5	3.6	7.5
SP	5.8	6.4	5.4	5.3	6.3	5.4	6.4	4.7	5	4	5.6	4.8
R	2.3	1.9	1.8	0	1.2	1.7	1	1.5	1	0	1.1	1.9
RB	10.4	10	3.7	9	9.6	3.9	15.3	13.4	7.6	18.4	4.2	5.9
SD	8.2	6.2	10.6	11.3	6.6	11.5	8.3	10.8	10.4	13.9	11.8	12.1
OT	7.3	8.9	6	8.5	9	6.9	4.9	7.7	7.7	6.4	7.6	7.2

Live Corals (LC), Partially Bleached (PB), Completely Bleached (CB), Soft Corals (SC), Dead Corals (DC), Dead Coral with Algae (DCA), Sponges (SP), Rocky bottom (R), Rubble (RB), Sand (SD), Others (OT).

**Figure 3.** Dead coral and dead coral with algal cover at different study sites after bleaching (July–December 2010)

A, C, D – Coral bleached with algae cover

B- Dead corals

Table 3. Percentage of partially and completely bleached corals in 2010 at different survey sites in the Andaman and Nicobar Islands.

PERCENTAGE	SURVEY SITES IN 2010 (March–June)			
	North Andaman	Middle Andaman	South Andaman	Nicobar group
Partially Bleached Coral	17.2 %	13.1 %	17.3 %	12.2 %
Completely Bleached Coral	11.6 %	9.1 %	13.1 %	16 %

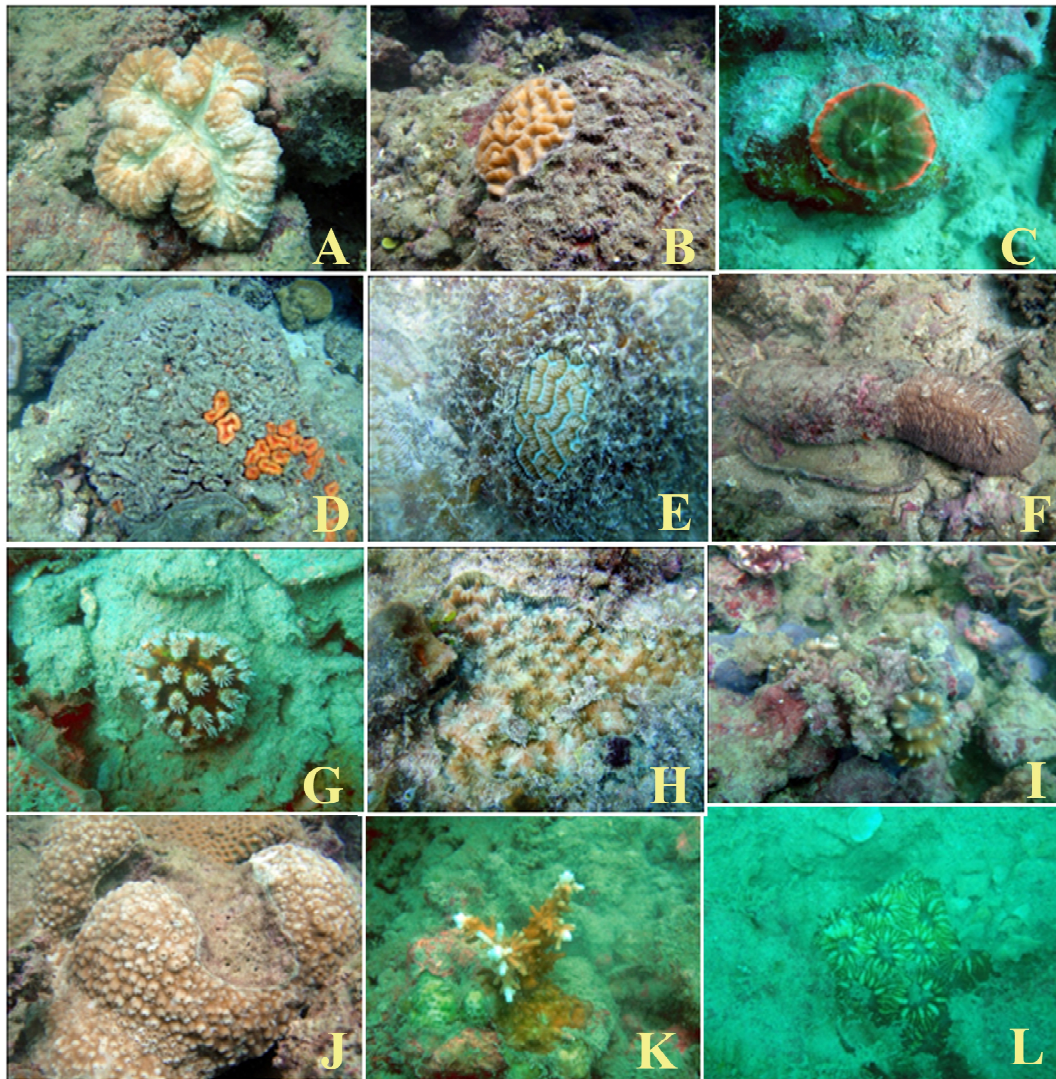


Figure 4. Recovered live corals at different study sites following bleaching (January–March 2011). [A - *Symphyllia valenciennesii* Milne Edwards and Haime, 1849, B - *Platygyra acuta* Veron, 2000, C - *Scolymia vitiensis* Bruggemann, 1877, D - *Lobophyllia hemprichii* (Ehrenberg, 1834), E - *Leptoria phrygia* (Ellis and Solander, 1786), F - *Herpolitha limax* Houttuyn, 1772, G - *Galaxea fascicularis* (Linnaeus, 1767), H - *Favites pentagona* (Esper, 1794), I - *Caulastrea furcata* Dana, 1846, J - *Astreopora myriophthalma* (Lamarck, 1816), K - *Acropora loripes* (Brook, 1892), L - *Acanthastrea regularis* Veron, 2002]

Table 4. Percentage coral recovery in 2011 at different survey sites in the Andaman and Nicobar Islands

PERCENTAGE	SURVEY SITES IN 2011 (January–March)			
	North Andaman	Middle Andaman	South Andaman	Nicobar group
Live Coral	17.5 %	14.3 %	14.8 %	15.4 %
Recovery	7 %	3.3 %	2.7 %	0.6 %

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